

## **Experiment 3**

### **CHARACTERISTICS OF AN ANTENNA**

## Experiment 3

### Objective:

- To study input impedance matching of an antenna.
- To study the radiation characteristics of an antenna.

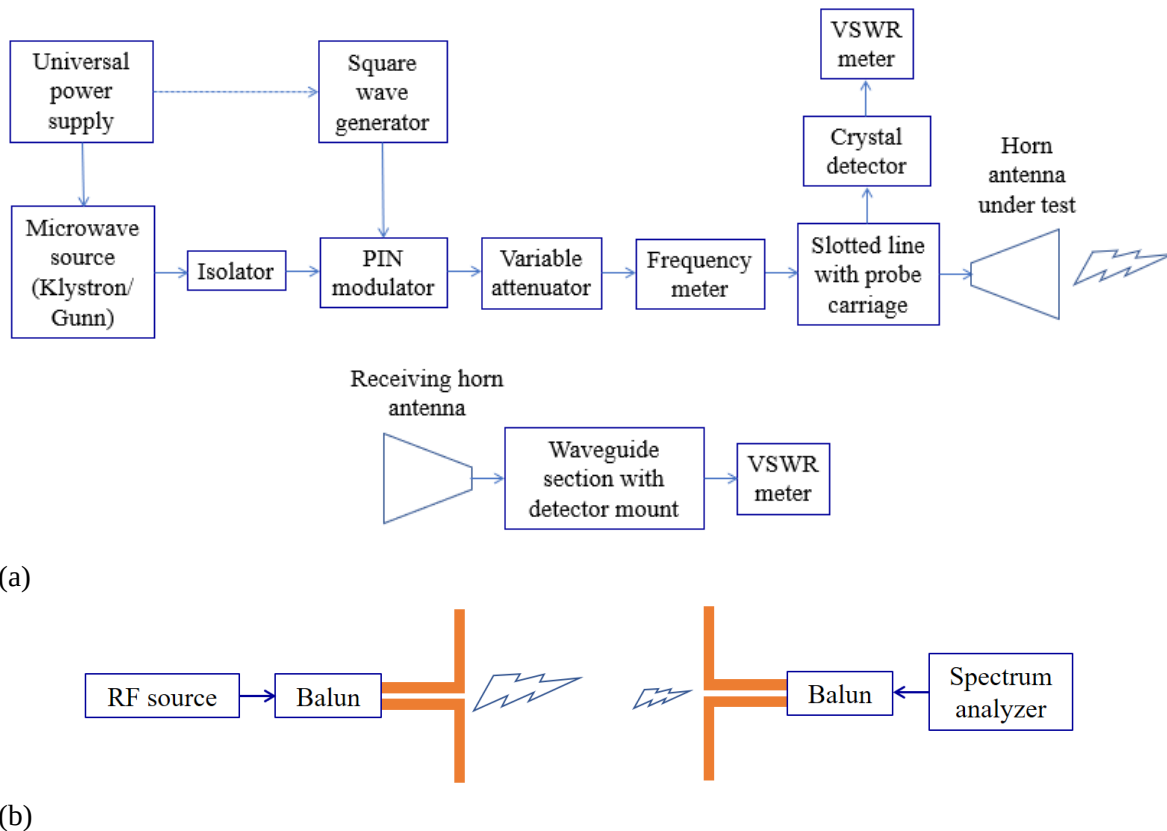


Fig. 1 Measurement set-up for the (a) horn antenna, and (b) dipole antenna.

**Theory:** Refer to 3<sup>rd</sup> design in the simulation study.

### Procedure: (horn antenna)

#### Input impedance matching:

1. Set up the experiment as shown in Fig. 1.
2. Connect the power supply to Klystron.
3. Adjust the Klystron repeller voltage for maximum reading in the VSWR meter.
4. Adjust the square wave modulation amplitude so that the detector output is maximum.
5. Tune the probe for maximum sensitivity by adjusting the tuning plunger on the slotted line.

6. Tune the wavemeter until a maximum fall in the square wave output amplitude is obtained. Note down the frequency reading.
7. Off-tune the wavemeter. Determine the VSWR of the antenna (see experiment 1).
8. Change the Klystron frequency by the tuning screw and repeat steps 3-7 for 8-10 frequency reading.
9. Convert VSWR to reflection coefficient (dB) and plot it versus frequency.

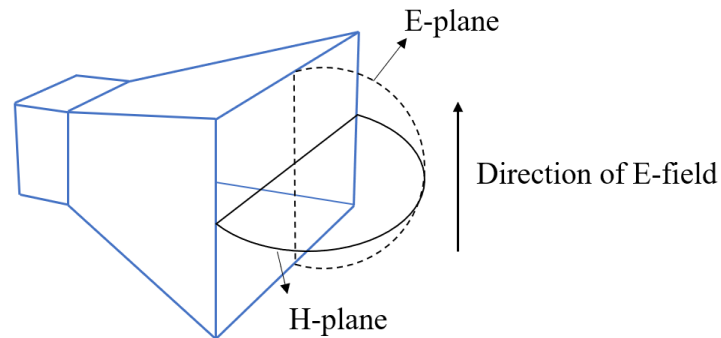


Fig. 2 E- and H-planes of the horn antenna under test.

#### **Radiation pattern:**

1. Align the receiving horn at  $0^\circ$  direction. Both the antenna under test and the receiving horn should be in the same polarization, say H-plane.
2. Set the Klystron at the mid-band frequency, say 10 GHz.
3. Follow steps 3-6 for maximum reading in the VSWR meter with the attenuator at maximum attenuation (at least 30 dB).
4. Connect the VSWR meter to the receiving horn antenna.
5. Tune the screw connected at the detector mount of the receiving antenna for maximum reading in VSWR meter.
6. Move the receiving horn to the right in  $5^\circ$  steps up to at least  $70^\circ$ . Note down the power reading for each step. (Received power is maximum at  $0^\circ$  direction. Use the corresponding VSWR meter reading as the reference. Now decrease attenuation to obtain the same reading for the new position of receiving horn. Thus, change in attenuation provides the change in received power in dB).
7. Plot relative received power versus angle.
8. Next, use  $90^\circ$  twist for both the antennas and repeat the same for E-plane.
9. From the plots, determine 3 dB beamwidths at the two planes.

#### **Boresight gain:**

1. Align the receiving horn at  $0^\circ$  direction.
2. Change the Klystron frequency by the tuning screw and repeat steps 3-7 in the first section.
3. Measure the relative received power at 8-10 frequency points.

4. Plot relative gain versus frequency.
5. Expression for approximate gain is

$$G = \frac{4\pi r}{\lambda_0} \sqrt{\frac{P_r}{P_t}} \text{ dB.}$$

- For the RF source and spectrum analyzer set up, the measurement process is quite straightforward. Set the power level and frequency on the RF source. Measure the received power at different angular positions using the spectrum analyzer. Repeat the measurements for at least two suggested frequencies and for both the E- and H-planes.
- Use the vector network analyzer to study the input impedance characteristics of a dipole antenna. Measure  $|S_{11}|$  and plot it.

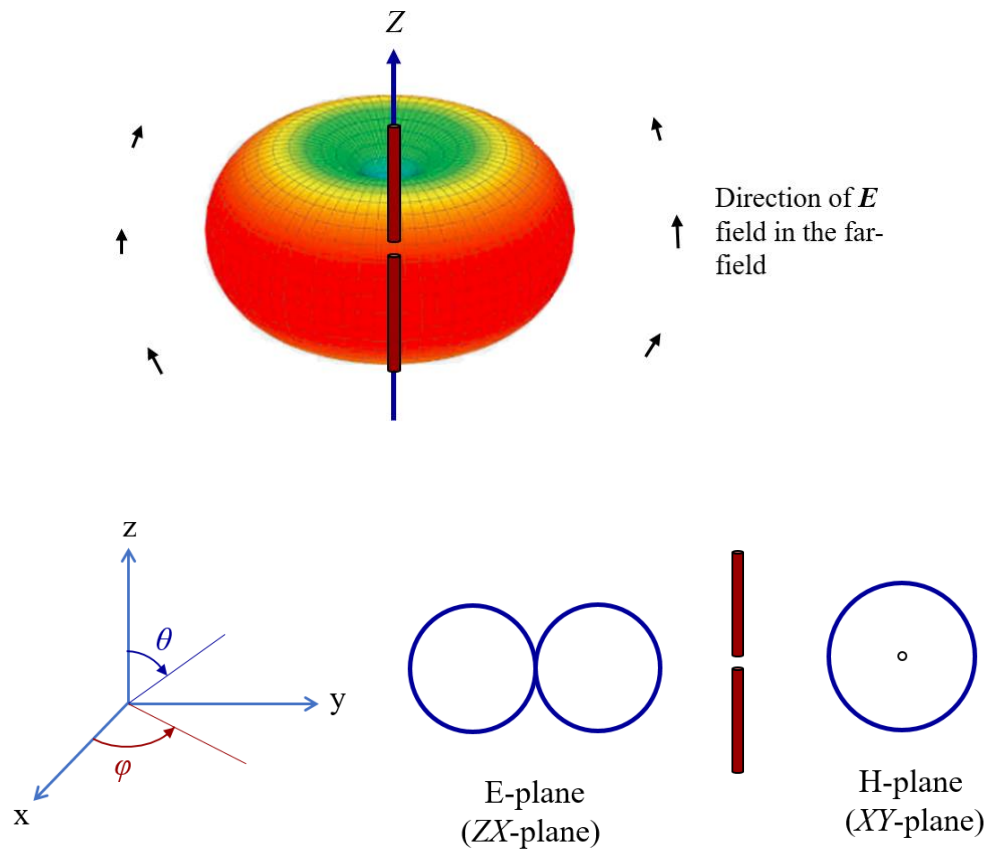


Fig. 3 Radiation patterns of a dipole antenna.